

Conditional sampling with $\mathcal{W}_1 \oplus \mathcal{W}_2$ proximals

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1 Conditional sampling

Let $(x, y) \in \mathbb{R}^{n+m}$ be distributed according to a joint density $\pi(x, y)$ with respect to Lebesgue measure, with marginal density $\pi_Y(y) = \int_{\mathbb{R}^n} \pi(x, y) dx$. Given only samples $\{(x_i, y_i)\}_{i=1}^I$ from the joint, we wish to produce samples from the conditional density $\pi(x | y) = \pi(x, y)/\pi_Y(y)$ using a generative flow. We learn a velocity field $v: [0, T] \times \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}^n$ that evolves a reference density $\mu_0(x)\pi_Y(y)$ to $\pi(x | y)\pi_Y(y) = \pi(x, y)$, where μ_0 is a reference density on $\mathbb{R}^n$ (e.g., a standard Gaussian, or a prior density for Bayesian inverse problems).

We approach this problem using $\mathcal{W}_1 \oplus \mathcal{W}_2$ proximal generative flows, which are better suited for distributions supported on low-dimensional manifolds. In particular, we use a Lipschitz-regularized f -divergence as the terminal cost. Denoting by Γ the class of 1-Lipschitz functions on $\mathbb{R}^{n+m}$, the joint formulation reads

$$\inf_{\rho, v} \left\{ \mathcal{D}_f^\Gamma(\rho_T \| \pi) + \frac{\lambda}{2} \int_0^T \int_{\mathbb{R}^{n+m}} \|v(t, x, y)\|^2 \rho(t, x, y) dx dy dt \right\}, \quad (1)$$

where $\rho_T := \rho(T, \cdot, \cdot)$ is the terminal-time density, and the infimum is subject to the continuity equation

$$\partial_t \rho + \nabla_x \cdot (v \rho) = 0, \quad \rho(0, x, y) = \mu_0(x) \pi_Y(y). \quad (2)$$

The kinetic energy term arises from a Benamou–Brenier reformulation of the $\mathcal{W}_2^2$ proximal penalty. Expanding the Lipschitz-regularized f -divergence via its variational representation gives

$$\inf_{\rho, v} \left\{ \sup_{\phi \in \Gamma} [\mathbb{E}_{\rho_T}[\phi(x, y)] - \mathbb{E}_\pi[f^* \circ \phi(x, y)]] + \frac{\lambda}{2} \int_0^T \int_{\mathbb{R}^{n+m}} \|v(t, x, y)\|^2 \rho(t, x, y) dx dy dt \right\}. \quad (3)$$

Since the initial density factorizes as $\mu_0(x)\pi_Y(y)$ and the velocity field acts only in the x -directions, the continuity equation (2) preserves the y -marginal: $\rho(t, x, y) = \rho(t, x | y)\pi_Y(y)$ for all $t \in [0, T]$, where $\rho(t, \cdot | y)$ evolves from μ_0 toward $\pi(\cdot | y)$. We may therefore disintegrate the optimization over y to obtain

$$\inf_{\rho, v} \int_{\mathbb{R}^m} \left\{ \int_{\mathbb{R}^n} \sup_{\phi \in \Gamma} [\phi(x, y) \rho(T, x | y) - (f^* \circ \phi)(x, y) \pi(x | y)] dx + \frac{\lambda}{2} \int_0^T \int_{\mathbb{R}^n} \|v(t, x; y)\|^2 \rho(t, x | y) dx dt \right\} \pi_Y(y) dy. \quad (4)$$

Here $v(t, x; y)$ emphasizes that y enters the velocity field as a parameter. Then for any fixed y , the flow $\dot{x} = v(x, t; y)$ for $x(0) \sim \mu_0(x)$ will be such that $x(T) \sim \rho_T(x|y) \approx \pi(x|y)$.

This setup can be adapted to solving Bayesian inverse problems in an amortized way. Let x be an unknown parameter and y be an observation. Then the distribution $\pi(x, y)$ is the joint distribution over parameters and observations, which can be factorized as a product of the prior distribution $\pi_{pr}(x)$ and the likelihood $\ell(y|x)$. From the joint, we apply the conditional sampling algorithm outlined above to produce samples from the posterior $\pi_{post}(x | y) \propto \ell(y|x)\pi_{pr}(x)$. These joint samples provide both the training pairs and the marginal samples on y needed to evaluate (4). Setting $\mu_0 = \pi_{prior}$ incorporates prior information into the reference density.

2 Conditional gradient particles algorithm

We can set up a conditional Wasserstein gradient flow for the Lipschitz regularized f -divergence with training similar to how the terminal condition is computed in the $\mathcal{W}_1 \oplus \mathcal{W}_2$ flow. Recall Wasserstein gradient flows, for a given divergence $\mathcal{D}_f^\Gamma$ with respect to target $\pi(x, y)$, we have

$$\frac{\partial \rho_t(x, y)}{\partial t} = \nabla \cdot \left(\rho_t(x, y) \nabla \frac{\delta \mathcal{D}_f^\Gamma(\rho_t(x, y) \| \pi(x, y))}{\delta \rho} \right). \quad (5)$$

The variational derivative of the divergence is given by the maximizer of the variational form, that is,

$$\frac{\delta \mathcal{D}_f^\Gamma}{\delta \rho} = \phi^* = \arg \sup_{\phi \in \Gamma} [\mathbb{E}_{\rho_t}[\phi] - \mathbb{E}_\pi[(f^* \circ \phi)]]. \quad (6)$$

These dynamics can be simulated through particles (X_t, Y_t) distributed according to $\rho_t(x, y)$:

$$\frac{d}{dt} \begin{bmatrix} X_t \\ Y_t \end{bmatrix} = -\nabla_{x, y} \phi(X_t, Y_t) \quad (7)$$

Similar to the flow we choose the y -marginal of $\rho_t(x, y)$ to match $\pi(x, y)$ so that the particles coordinates are fixed along those marginals. That is, we should show that for fixed y

$$\dot{X}_t = -\nabla \phi(X_t, y) \quad (8)$$

samples from the conditional distribution $\rho_t(x|y)$, and that $\rho_t(x|y) \rightarrow \pi(x|y)$ as $t \rightarrow \infty$.

3 Application to Euler equations

Suppose we have a rigid body tumbling in the absence of gravity. Denote $\boldsymbol{\omega}(t) \in \mathbb{R}^3$ to be the angular velocity with respect to the inertial lab frame written in terms of the basis of the rotating body frame, $\mathbf{I}$ to be the inertia tensor with respect to the rotating body frame. The inertia tensor is an element of the space of symmetric positive definite matrices S_+^3 . Under torque-free dynamics, the Euler equations describe the evolution of the angular velocity over time

$$\begin{aligned} \mathbf{I} \dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times (\mathbf{I} \boldsymbol{\omega}) &= 0 \\ \boldsymbol{\omega}(0) &= \boldsymbol{\omega}_0 \end{aligned} \quad (9)$$

Here, $\times$ denotes the vector cross-product. The attitude of a rigid body is a rotation matrix $\mathbf{R}(t) \in SO(3)$ that maps body frame vectors to lab frame vectors, which encodes information about the orientation of the body with respect to the lab frame. Each component of the rotation matrix $r_i(t)$ evolves according to $\dot{r}_i = (R\boldsymbol{\omega}) \times r_i$, which can be written in terms of a matrix multiplication as

$$\dot{\mathbf{R}} = \mathbf{R} \hat{\boldsymbol{\omega}}, \quad \hat{\boldsymbol{\omega}} = \begin{bmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{bmatrix}. \quad (10)$$

3.1 Bayesian inverse problem

Suppose we only have noisy measurements of $\mathbf{R}$ with respect to some fixed set of vectors $\mathbf{b}_m \in \mathbb{R}^3$, $m = 1, \dots, M$, $M = 1, 2, 3$. That is we have a likelihood model of the form

$$y_{mn} = \mathbf{y}_m(t_n) = \mathbf{R}(t_n) \mathbf{b}_m + \xi_{mn}, \quad (11)$$

where $\xi_{mn} \sim \mathcal{N}(0, \sigma^2 I_3)$. We wish to infer the parameters $\theta = (\mathbf{R}(0), \boldsymbol{\omega}(0), \mathbf{I})$ given observations y_{mn} at discrete time steps. Each evaluation of the likelihood for a given θ requires an ODE solve of the Euler equations. This can be prohibitively expensive for traditional Bayesian computation methods such as Markov

chain Monte Carlo; we will address this through simulation-based inference and conditional generative modeling which only requires samples of the likelihood.

We set the prior on $\mathbf{R}_0 \sim \pi(\mathbf{R})$ to be a uniform distribution on $SO(3)$ and $\boldsymbol{\omega}_0 \sim \pi(\boldsymbol{\omega}_0)$. For the inertia tensor, a prior is the inverse Wishart distribution, which is a distribution on symmetric positive definite matrices. The structure of this prior yields a posterior distribution that is supported on $SO(3) \times \mathbb{R}^3 \times S_+^3 \subset \mathbb{R}^{18}$, a 12-dimensional manifold embedded in 18 dimensions. The low dimensional nature is due the fact that $SO(3)$ is the space of orthogonal matrices with determinant equal to 1, which is fundamentally 3-dimensional.

To produce the samples to train the conditional generative model, produce K samples from the prior $\theta^{(k)} \sim \pi(\theta)$ and use the same parameters to produce the observational data $y_{1:M,1:N}^{(k)} \sim \pi(y_{1:M,1:N}|\theta^{(k)})$. We first train a $\mathcal{W}_1 \oplus \mathcal{W}_2$ flow.

4 Application related to Canonical Ensembles in Stat Mech

... or in Geometric Methods for Molecular Dynamics (preserving the canonical ensemble)—see book by Leimkhuler and Reich for examples.